

Test 7: Decay slope (log-log regression)

Verdict: PASS | Generated: 2026-09-02 13:49:32 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Large-T decay of b(N): formula family (slope??0.5), verified via linear regression with R?.

METHOD: Large-T decay of b(N): formula family (slope??0.5), verified via linear regression with R?.

Result

Test 7 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

`plots/plot\_01` (.png 600dpi / .svg / .pdf)

Plots

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 7 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(no log_comp entries)

FULL COMPUTATION LOG

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CONSOLE CAPTURE

TEST 7 (secondary, HARDCORE): Objection 3: Large-T Decay Rate ? deep slope validation

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D1 slope : -0.1504, 95%CI [-0.1594, -0.1414] (width 0.0179), R?=0.9954 (limits 0.1 / 0.9)

D2 stability : slope_lo=-0.1752, slope_hi=-0.1385, |?|=0.0367 (limit 0.15)

D3 walk-fwd : median err 0.05%, max err 0.09% (limits 2%/5%)

D4 bootstrap : slope=-0.1504 inside boot 95%CI [-0.1621, -0.1378]? YES

??

D1 slope CI PASS ? width 0.0179, R? 0.9954

D2 stability PASS ? |?slope| = 0.0367

D3 walk-forward PASS ? median 0.05%, max 0.09%

D4 boot vs CI PASS ? CI [-0.1621, -0.1378]

Test 7 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS